MCC Entrance Test, August 2026 — Day 2

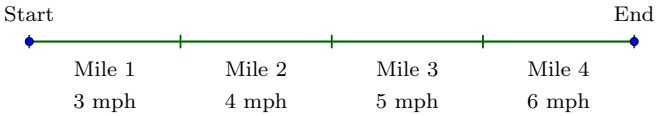
Level T3

Test Information. Level T3 — 150 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra, Combinatorics (×2), Geometry, Number Theory (×2).
Rules: show your work (partial credit rewards genuine progress, but claims still need real justification at this level); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts). Every cell of an $n \times n$ grid is colored with one of three colors – Red, Green, or Blue. Find the smallest n for which, no matter how the colors are assigned, some row or some column is forced to contain three cells of the same color.

Problem 2 (10 pts). A four-digit number $\overline{aabb}$ – whose first two digits are equal and whose last two digits are equal – is a perfect square. Find this number.

Problem 3 (15 pts). A runner completes a four-mile course. She covers the first mile at 3 mph, the second mile at 4 mph, the third mile at 5 mph, and the fourth mile at 6 mph.



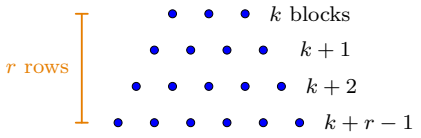
(a) Find the total time, in hours, the run took. [6 pts]
(b) Using your answer to part (a), find the runner's average speed for the entire course, expressed as a fraction m/n in lowest terms. Report $m+n$. [9 pts]

Problem 4 (15 pts). Let ABC be a triangle with $\angle A = 60°$. Points K, M, N lie on sides AB, BC, CA respectively, so that $BK = KM = MN = NC$. Given that $AN = 2\,AK$,

(a) Find $\angle KMN$. [4 pts]
(b) Find $\angle B$ and $\angle C$. [11 pts]

Problem 5 (25 pts).

A *stack* is an arrangement of blocks in rows, where every row (after the top one) has exactly one more block than the row directly above it. A positive integer n is called a *stack-number* if some stack uses exactly n blocks in total, with at least two rows and at least two blocks in the top row.



(a) For a stack with r rows ($r \geq 2$) and k blocks in the top row ($k \geq 2$), find a formula for the total number of blocks n, in terms of r and k. Writing $s = 2k + r - 1$, use your formula to show that $2n = r \cdot s$, and then prove that r and s are always positive integers of opposite parity, with $s > r$. [6 pts]
(b) Using part (a), show that no stack-number is ever a power of 2. [3 pts]
(c) Using parts (a)–(b), determine how many stack-numbers there are less than 100. [16 pts]

Problem 6 (25 pts). A positive integer m satisfies $m^{1003} \equiv 19 \pmod{77}$.

(a) Find the remainder when m is divided by 11. [12 pts]
(b) Find the remainder when m is divided by 7. [3 pts]
(c) Using parts (a)–(b), find the remainder when m is divided by 77. [10 pts]